

SIH26018 - Intelligent Land Record Digitization and Validation System

Ministry of Rural Development

Problem Statement

- Land records exist in old paper documents, maps, registers, spreadsheets etc.
- These records are not in digital format and are difficult to search, access and manage.
- Data is incomplete, inconsistent and contains errors.
- Manual verification is time consuming and prone to human mistakes.
- Lack of a system to validate extracted information and detect inconsistencies.

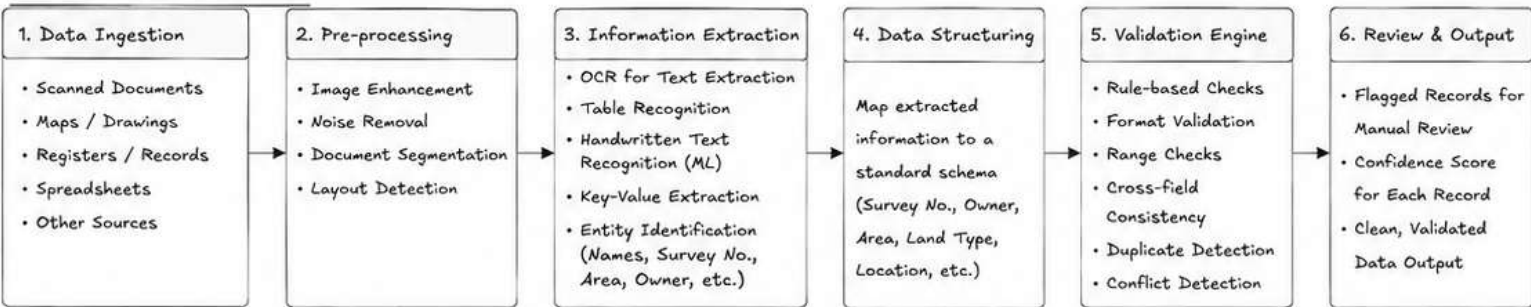
Our Solution (High Level)

An AI-powered system that digitizes legacy land records from multiple formats and validates the extracted data for accuracy, completeness and consistency.

It will:

1. Convert paper documents, maps and registers into structured digital data.
2. Extract key information automatically using AI/ML.
3. Validate the extracted data using rules, constraints and cross-document checks.
4. Flag errors, missing values and inconsistencies for review.
5. Store clean, validated land records in a standardized digital format for further use.

Solution Flow (Roadmap)



Technologies to be Used

- AI/ML: OCR (Tesseract, EasyOCR), Layout Detection (DocLayout-YOLO), Table Recognition, Handwriting Recognition (CRNN/Transformer)
- Data Processing: Python, OpenCV, Pandas, Regular Expressions
- Validation: Rule Engine, Business Rules, Data Quality Framework
- Database: PostgreSQL / MySQL (Structured Data Storage)
- Backend: FastAPI / Django
- Frontend: React / HTML, CSS, JavaScript (Dashboard for Review & Management)
- Others: Elasticsearch (Search), Docker (Deployment)

Challenges / Next Problems	How We Handle Them	Outcome
1. Poor quality / unclear documents	→ Image enhancement, advanced OCR models, human-in-the-loop for low confidence cases	• Accurate and validated digital land records • Reduced manual effort and time • Improved data quality and consistency • Ready-to-use structured data for land governance, harmonization and decision making
2. Handwritten and multi-language text	→ Handwriting recognition models, language detection and translation	
3. Incomplete or missing information	→ Missing value detection, confidence scoring, mark for follow-up	
4. Conflicting information in records	→ Cross-document validation, conflict detection rules, flag for manual review	
5. Large volume of data	→ Scalable architecture, batch processing, parallel pipelines	
6. Human verification still required	→ User dashboard for review, edit, approve and feedback loop for learning	

Future Improvements

- Integration with GIS for spatial validation
- Linking with external databases (Revenue, Survey, Cadastral)
- Continuous learning from user feedback
- Mobile capture app for new records
- API for other government systems

SIH26013 – Automated Integration and Intelligent Harmonization of Multi-source Geospatial Data for Urban Land Record Management

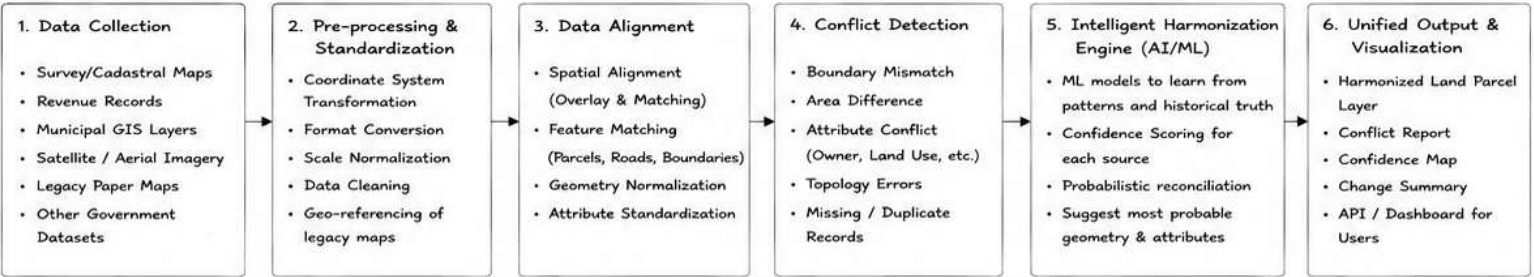
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Problem that arose in SIH26013 (Validation Problem):
We cannot confirm which land record is true and which one is false because different sources give conflicting information about the same land.

- Problem Statement**
- Land information comes from multiple sources (survey maps, revenue records, municipal GIS, cadastral maps, satellite images, etc.).
 - Each source uses different formats, scales, coordinate systems and data standards.
 - The same land parcel often has different boundaries, areas or ownership details in different sources.
 - Manual comparison of maps and records is slow, error-prone and not scalable.
 - There is no automated system to integrate all sources and identify conflicts intelligently.

- Our Solution**
- An AI-powered system that automatically integrates and harmonizes multi-source geospatial data to create a single, consistent and trusted view of land parcels.
- It will:
1. Bring all land data sources to a common standard.
 2. Detect conflicts and inconsistencies automatically.
 3. Use AI/ML to suggest the most probable (not absolute) version with a confidence score.
 4. Provide a unified, harmonized land dataset for further use in governance and decision making.

Solution Flow (Roadmap)



- Key Innovations by Us (X-Factor)**
- **AI-based Probabilistic Harmonization:** Instead of rule-based matching only, we use ML models to learn from historical verified data and estimate the most credible version with a confidence score.
 - **Multi-level Conflict Intelligence:** Not just detecting conflicts, but classifying the type, cause and severity of conflict (boundary shift, data entry error, projection issue, real-world change, etc.).
 - **Change Timeline Reconstruction:** Using time-series satellite imagery and historical maps, we estimate when a change in boundary or land use actually happened.
 - **Explainable Results:** For every decision, system shows "Why" a particular source is trusted more (evidence view).
 - **Adaptive Learning:** As more verified data comes, the model improves continuously.

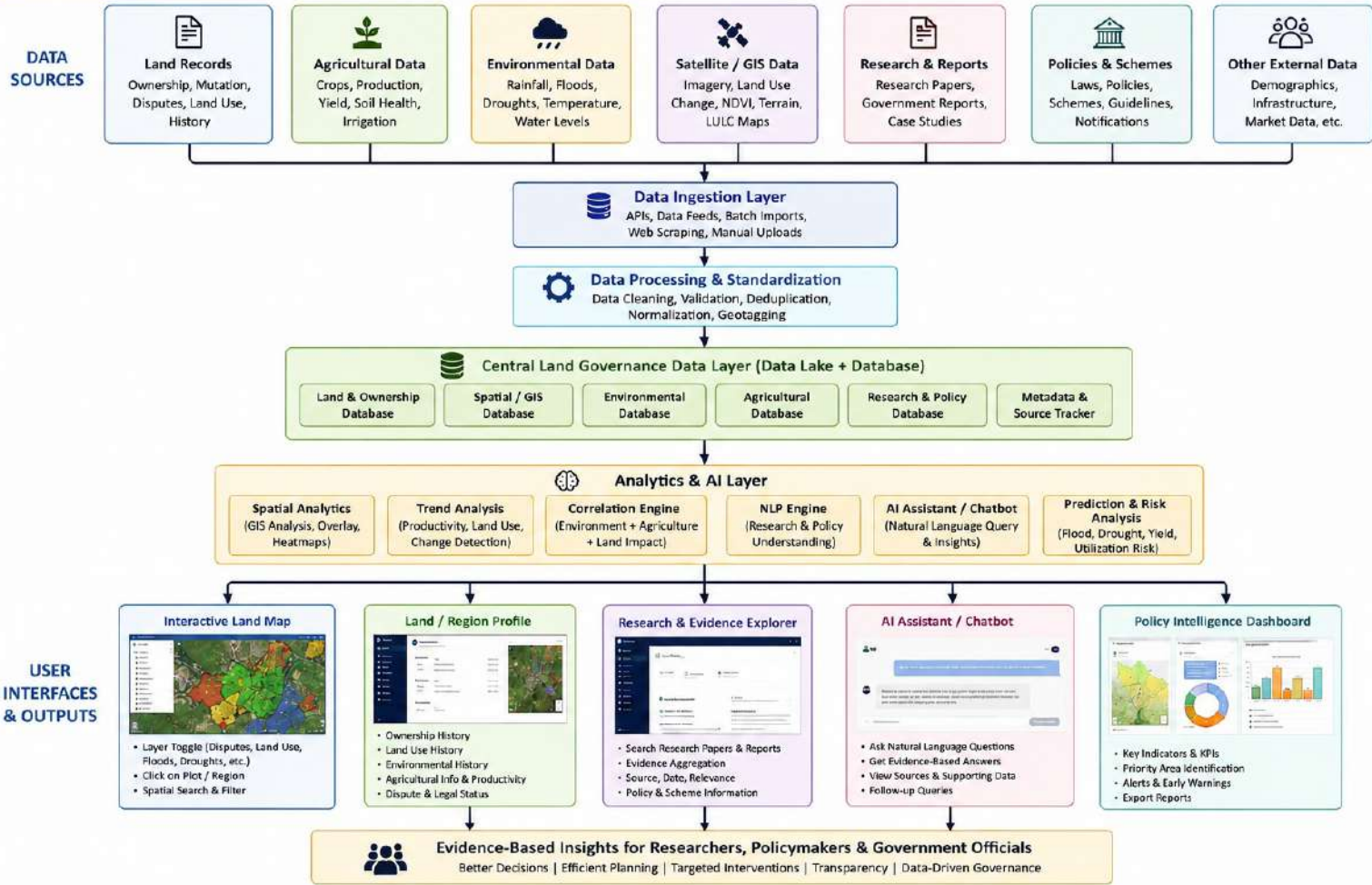
Challenges / Next Problems	How We Handle Them	Outcome
1. Different coordinate systems and projections	→ Automated CRS detection and transformation to a common standard (EPSG based).	All data aligned in the same space.
2. Boundary and area mismatches	→ Spatial overlay + ML based boundary adjustment + confidence scoring.	Most probable boundary with confidence %.
3. Conflicting attribute information	→ Source reliability scoring + historical verification + majority pattern learning.	Most credible attribute suggested.
4. Missing or incomplete data	→ Gap detection + data imputation (where possible) + flag for manual review.	Complete and usable dataset.
5. Large volume of heterogeneous data	→ Scalable Geo-data pipeline using cloud + spatial databases.	Faster processing and scalability.
6. Deciding truth vs false	→ Probabilistic approach (not absolute truth) + human-in-loop verification.	Reliable, transparent and explainable output.

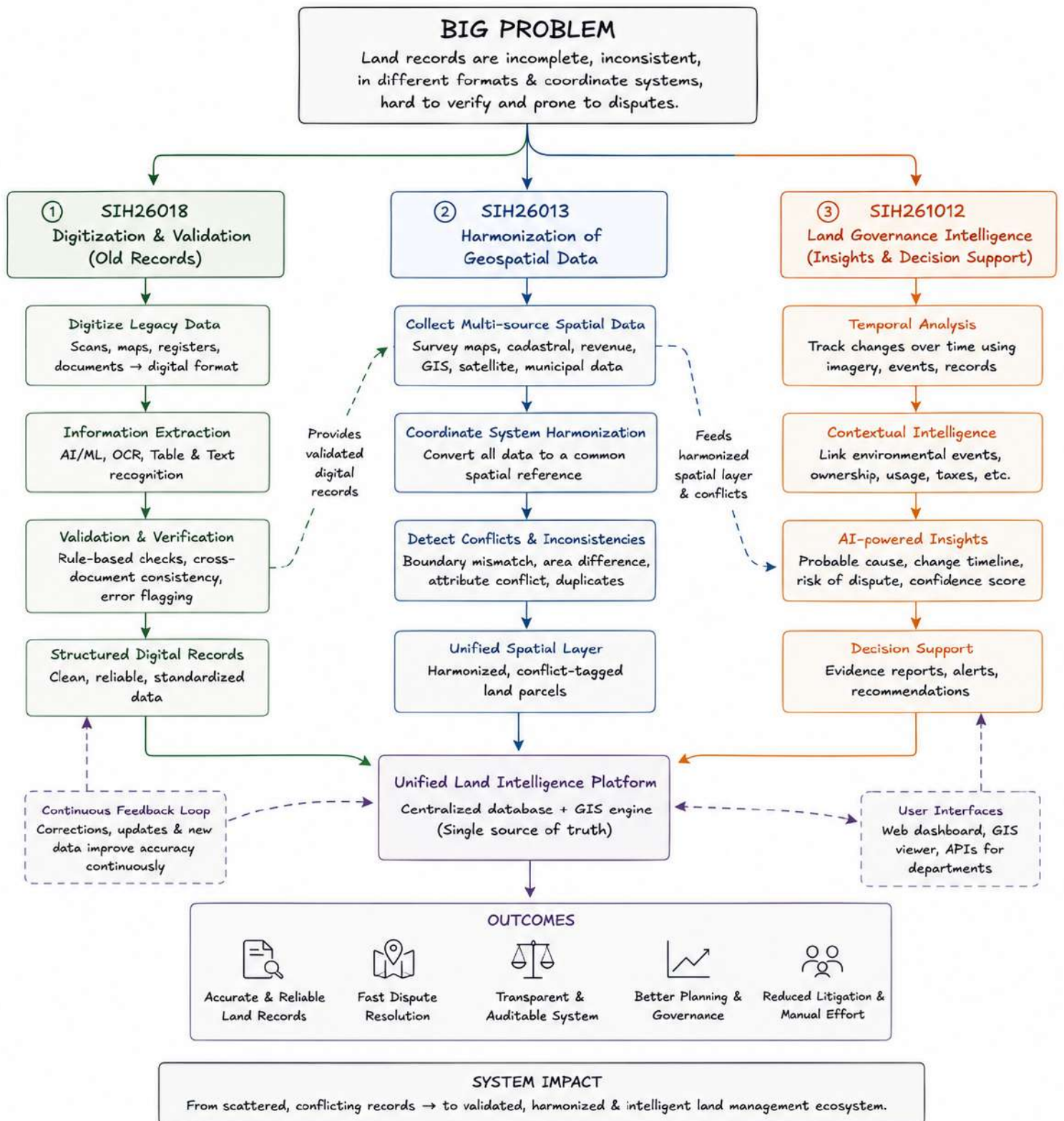
- Technologies to be Used**
- GIS: QGIS, ArcGIS, PostGIS
 - Programming: Python
 - Geospatial Libraries: GDAL/OGR, GeoPandas, Shapely, Rasterio
 - AI/ML: Scikit-learn, XGBoost, TensorFlow
 - Data Processing: Pandas, NumPy
 - Database: PostGIS, PostgreSQL
 - Backend: FastAPI / Django
 - Frontend: React / Leaflet / Mapbox
 - Cloud: AWS / GCP
 - Others: Docker, Elasticsearch

SIH26019

NATIONAL LAND GOVERNANCE INTELLIGENCE PLATFORM

System Flow Map





SIH26104 – Telephony-Resilient Voice Deepfake & Impersonation Detector

Ministry of Home Affairs

THE PROBLEM

AI voice generators can clone any person's voice using just a 3-second sample. Scammers use these cloned voices for impersonation scams, "digital arrests" and financial fraud over phone calls.

Most AI voice detectors work only on clean studio audio (.wav). They fail on real phone calls (VoLTE/GSM) because of 8kHz downsampling, heavy compression and background noise which destroy the features those models depend on.

OUR SOLUTION

An end-to-end, low-latency audio authentication pipeline that analyzes live incoming audio streams and flags synthetic voice clones within 3 seconds, designed to work under real telephony conditions.

CORE INNOVATIONS

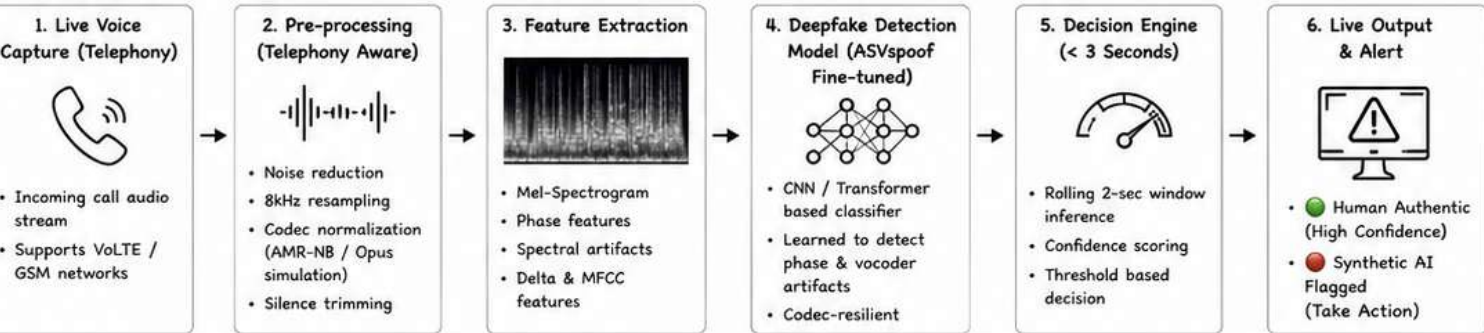
- 1

Phase & Vocoder Artifact Spotting (Bio-Physical Analysis)
 - Human speech is produced by physical vocal cord vibrations, causing natural micro pitch variations, breath pauses and phase patterns.
 - AI voice engines generate audio frame-by-frame, leaving invisible "stitching marks" (phase inconsistencies, spectral smoothing).
 - We analyze Mel-Spectrograms to detect these micro-stitching artifacts that humans cannot hear.
- 2

Codec-Resilient Audio Model Training (Telephony Defense)
 - Model fine-tuned on ASVspoof dataset with augmentation that simulates mobile network degradation (Opus, AMR-NB codecs, 8kHz downsampling, packet loss, noise).
 - Ensures high detection accuracy even during real cellular phone calls.
- 3

Sub-3-Second Rolling Window Streaming Inference
 - Uses a 2-second sliding audio buffer to analyze voice streams in real time.
 - Provides a visual warning (🟢 Human Authentic / 🔴 Synthetic AI Flagged) within 3 seconds of the caller speaking, enabling action during the call.

SOLUTION WORKFLOW



TECHNOLOGIES USED

- Programming: Python
- Audio Processing: Librosa, Torchaudio, NumPy
- Deep Learning: PyTorch
- Model: CNN / ResNet / ECAPA-TDNN (fine-tuned on ASVspoof)
- Deployment: FastAPI (Backend API)
- Frontend: HTML, CSS, JavaScript (Web UI)
- Visualization: Plotly.js / Chart.js (Waveform & Scores)
- Others: Docker, Nginx (Deployment)

EXPECTED OUTCOME

- Detects AI voice clones during real phone calls in under 3 seconds.
- Maintains high accuracy even under heavy compression and noise.
- Helps prevent fraud, impersonation scams and "digital arrests".
- Provides real-time alerts to users or call centers.
- Scalable for integration with telecom / law enforcement systems.

DEMO SETUP (HACKATHON)

- Frontend : Web UI showing live waveform, confidence score and real/fake status.
- Backend : Python API (FastAPI) using PyTorch & torchaudio to process incoming audio, extract features, run inference and stream results to the UI.